HACKATHON 2026 · PROTOTYPE SUBMISSION

SevaSetu

Bridge to Hope

An AI-powered platform using Google Gemini and real-time geospatial intelligence to automate crisis triage, prioritize community needs, and coordinate volunteers — ensuring the right help reaches the right place at the right time.

AI

Urgency Scoring

Maps

Geospatial Heatmap

GCP

Cloud Native

- Google Cloud Run
- Gemini AI 1.5 Flash
- Google Maps Platform
- Firebase Auth

THE PROBLEM

Crisis Response is Broken

Delayed Triage

Authorities manually analyse reports, wasting the critical
'Golden Hour' while situations worsen on the ground.

No Location Intelligence

Volunteers are deployed arbitrarily — without knowing
which zones are most critically affected in real-time.

Resource Wastage

Duplicate efforts in some areas while high-urgency needs in other zones go completely ignored.

"During the 2023 Flood Response in Assam, 40% of NGO resources were misdirected due to lack of a central prioritization system — costing 72+ hours of critical response time."

OUR SOLUTION

SevaSetu: End-to-End AI Coordination

AI-Driven Urgency Scoring

Google Gemini 1.5 Flash analyses every report and assigns an urgency score 0–100 instantly. Medical, water, food, shelter — prioritised in seconds.

Geospatial Crisis Heatmap

Google Maps displays a real-time heatmap of active need reports so coordinators see exactly where to deploy volunteers for maximum impact.

Citizen-Powered Reporting

Community members submit geo-tagged reports with photos (stored in GCS) via a mobile-friendly UI with Places Autocomplete for accurate location input.

Instant Volunteer Alerts

Firebase Cloud Messaging (FCM) pushes real-time task assignments to volunteers the moment a high-urgency need is identified in their area.

TECHNICAL ARCHITECTURE

Built on Google Cloud

Community
React + Vite

→

Firebase Auth
Google Sign-In

→

Cloud Run API
Node.js / Express

→

Gemini AI
1.5 Flash Model

Prisma ORM
Structured DB

Cloud Storage
Photo Uploads

Maps Platform
Heatmap + Places

FCM Alerts
Push Notifications

Region: asia-south1 (Mumbai)

Containerized via Docker

Cloud Build CI/CD

Firebase Hosting (Frontend)

🔧 GOOGLE SERVICES INTEGRATION

Powered by Google Ecosystem

🤖

Google Gemini AI 1.5 Flash

Auto-analyses each report in Hindi/English. Assigns urgency score, priority label (CRITICAL / HIGH / MEDIUM / LOW) and a one-line reason for coordinators.

PRIMARY AI MODEL

🗺️

Google Maps Platform

Interactive crisis heatmap, marker clusters, and Places Autocomplete for accurate geo-tagged location input during report submission by community members.

MAPS JS API + PLACES API

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FIREB

🚀 LIVE PROTOTYPE

Fully Deployed & Production Ready

🌐 **LIVE FRONTEND**

https://seva-setu-74bcc.web.app

⚡ **LIVE API (Cloud Run)**

https://sevasetu-api-1032768844799.asia-south1.run.app

💻 **GITHUB REPOSITORY**

https://github.com/AnkitG791m/seva-setu

User Flow Walkthrough

1. Google Login → Role Selection
2. Community member submits Need report with photo + location
3. Gemini AI scores it instantly (e.g. CRITICAL: 87/100)
4. Maps heatmap updates — coordinator assigns nearest volunteer
5. FCM Push notification sent to volunteer in real-time

SevaSetu App

🌉

✓ Google Login Active

🤖 AI: Scoring Reports...

🔴 CRITICAL · Urgency: 91/100

🔭 FUTURE ROADMAP

What's Next for SevaSetu

Phase 2: Predictive Hotspot Detection

Use Vertex AI to analyse historical crisis data and predict vulnerable zones before incidents occur — enabling proactive resource pre-positioning before disasters strike.

Phase 3: Offline-First Mode

SMS/USSD gateway integration so field workers in remote areas with no internet can still submit reports and receive task assignments via basic mobile phones.

Phase 4: Drone Zone Mapping

Automated drone deployment to high-urgency zones identified by SevaSetu AI, enabling aerial surveys, supply drops, and real-time visual confirmation of crisis situations.

📊 Scale Targets

Reports / day	100,000+
Active Volunteers	50,000+
Response Time	< 15 mins
States Covered	All 28 States

Building a Safer India

SevaSetu is more than a platform — it's a commitment to ensuring that no community need goes unheard, no volunteer is misdirected, and every crisis is met with the precision of artificial intelligence.

Live App
seva-setu-74bcc.web.app

GitHub
AnkitG791m/seva-setu

API
Cloud Run · asia-south1

● Deployed on Google Cloud ● Powered by Gemini AI ● Firebase + Maps Platform